

Fraud Detection in Online Transaction using Hybrid CNN and XGBOOST

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Abstract— Online transactions have become an essential part of our daily life; this makes such transactions more probable to result in fraud. The users card details have been abused by fraudsters and online transactions are made through them. The scammers are always looking for new tools and ways to carry out their fraudulent activities. Machine learning algorithms have the capacity to develop and identify fraud patterns that were not previously observed. In this study, datasets of financial transactions are applied in various machine learning algorithms to solve the problem of transaction fraud. These techniques are based on the hybridization form of Convolution Neural Network and Support Vector Machine (CNN-SVM) and Extreme Gradient Boosting algorithm (XGBOOST). The SVM classifier was used to replace the CNN architecture model's final last output layer in order to create the hybrid CNN-SVM model architecture. An enhanced variation of the gradient-boosting strategy is called XGBOOST. It enhances the system's performance by employing ensemble approaches. To address an unbalanced class distribution, the current classification models are modified using ensemble approaches. The XGBoost model and the hybrid model are contrasted. We demonstrate that our suggested model can identify fraudulent transactions with a high degree of accuracy and a comparatively low quantity of false positives.

Keywords—Convolutional Neural Network (CNN), Fraud detection, Hybrid model, online transactions, Support Vector Machine (SVM), Extreme gradient Boosting (XGBOOST).

I. INTRODUCTION

The environment in which we currently live is changing quickly, and digital payment methods are one of them. Transaction volumes for credit card and payment firms have increased dramatically. Financial fraud in modern payment systems is increasing quickly along with these transactions. A highly accurate fraud detection system should be able to identify fraudulent transactions. It ought to arrive as no surprise that nearly three-quarters of the companies believe digital payment fraud to be a significant issue, given the amount of data around this issue worldwide. It can be alarming and frustrating for customers to have their credit card information stolen. The victims of payment method fraud typically need to cancel their cards and cope with the fallout for a period of two days. Through the internet fraud is a major expense for online retailers and, according to 44 percent of finance professionals, it's their top issue. In the event that consumers become victims of fraud, they forfeit the bought goods. The retailer is then billed for the bogus purchase a process known as a chargeback after the buyer notifies the bank of the crime. Additionally, the seller must

reimburse making payments provider for reversal expenses. This process involves leveraging advanced technologies and algorithms to identify and mitigate fraudulent activities during online transactions. The primary goal is to ensure the integrity of financial transactions and protect users' sensitive information. One of the key components of online transaction fraud detection is machine learning. Through the analysis of massive volumes of transaction history, predictive algorithms are able to identify trends indicative of normal behaviour and detect anomalies that may signal fraudulent activity. These models continuously evolve to adapt to new and sophisticated fraud techniques.

A comparative analysis of the outcomes of Extreme Gradient Boosting (XGBOOST) and hybrid CNN-SVM in this work. Two important algorithms are utilized in the hybrid model's implementation. Due to its efficacy on datasets with numerous features particularly when the total amount of features increased and the amount of data points CNN is hybridized with support vector machines. To build a model with the best type of parameters and a highest performance function in detecting fraud. Hyperopt is used to optimize the XGBoost classifier. There are distinct application cases for the two algorithms. Compared to other methods, the final model will have a greater accuracy rate. When it comes to operating time and fraud transaction prediction accuracy, our system offers two key advantages.

II. LITERATURE REVIEW

“Tamiru Melese, Assaye Walelign and Abdu Mohammed” [1] In this study, a hybridization function model is presented in this work to identify credit card transaction fraud. It employs actual, publicly available credit card transaction data. An SVM classifier was used to replace the CNN model's final output layer in order to create the hybrid model's architecture. The experimental results demonstrate that, when it comes to prognosis whether it comes under the crime of detecting fraud and considering it an fraudulent activities, our suggested strategy outperforms the fully connected convolution neural network predicts the accuracy. “Xinxin Zhou, Xiaobo Zhang, Lizhi Wang, Pengwei Wang” [2] In this paper, the area of online transactions, this research suggests a detecting model of fraud based on convolution neural networks. This model's benefit is that are used as derivative, safeguarding online transaction data norm as input format. The experimental consequence shows that, when

validated against a digital payment sample from the bank, the approach has exceptional identification of fraud efficacy.

This approach reduces the amount of time needed to calculate the derived variables when compared to the current CNN model, which speeds up and simplifies the model's design and tuning process. Additionally, the accuracy with which scams are identified has increased.

“Yixuan Zhang, Jialiang Tong, Ziyil Wang, Fengqiang Gao” [3] This study proposes a fraud detection algorithm based on XGBoost. In this case, the problem of imbalanced identifiers was resolved besides to using in particular function. Increasing the database set in a balanced manner is accomplished with this strategy. The label encoding algorithm is used to encode categorical data for features that are classified as such. For the categorization, the extremely useful and applicable XGBoost algorithm was utilized. Without sampling, the suggested XGBoost can achieve improved accuracy for imbalanced data.

“Sheo Kumar, Vinit Kumar Gunjan, Mohd Dilshad Ansari, Rashmi Pathak” [4] In this paper, Support Vector Machine algorithm (SVM) classification is taken in the suggested approach to identify frauds. This model is evaluated against state-of-the-art classifiers, including SVM and other machine learning algorithms. Using credit card information of the users, this fraud detection system can manage and stop the attacker's transactions. The outcome demonstrates that the SVM kernel performs better in terms of accuracy.

“Krishna Kumar Mohbey, Mohammad Zubair Khan, Ajay Indian” [5] In this paper, to ascertain whether or not a credit card transaction comes under fraudulent, this study compares the performance of existing conventional machine learning models with the ensemble technique-based XGBoost model. The risks of monetary losses and uncertainty that institutions face in their daily operations are greatly decreased by our study. An openly accessible dataset of credit cardholders is used for this. The ensemble model performs more well in precision, recall values, when XGBoost is used. Given that conventional models perform poorly when dealing with unbalanced data, the XGBoost model employs ensemble approaches to outperform other conventional models, increase performance, and achieve prediction accuracy of up to 96%. This research is important because of the exponential development in e-commerce activities due to their phenomenal expansion.

“Bocheng Liu, Xiang Chen and Kaizhi Yu” [6] This study produced multiple fraud detection techniques that used Highly Connected Neural Network and XGBoost, as well as a digital transaction fraudulent detection method using on XGBoost in predictor. The previous method integrated twin neural network designs that had distinct crossover entropy loss parameters, and it is an easy and fast process to generate the resulting model. The XGBoost classifier was optimized using Hyperopt by the latter technique, allowing for the construction of a model with optimal parameters and good fraud detection performance. We made the decision to create an XGBoost is used in fraud in electronic transaction detection systems to ensure high accuracy for detection.

Using Hyperopt, this technique builds the best-performing XGBoost predictor variables. This classifier's AUC may reach 0.969, which significantly enhances network transaction fraud detection performance.

III. EXISTING SYSTEM

Several machine learning techniques are used in current systems to detect online transaction fraud and predict credit card fraud. The person model has a special function in identifying dishonest behaviour. In order to create various convolutional patterns, CNN builds an layer of input characteristics processing that reorganizes raw transaction characteristics. The fact that different configurations that come into the inversion core may result in various characteristics makes it significant. The network is composed of four layers: first is feature-successiveness, second convolutional layer, pooling method, and fully connected layers. Excellent fraud detection results are obtained in this model [8]. SVMs are a promising method for estimating functions in regression problems or for learning distinct functions in pattern recognition tasks. Optimal hyper planes are unsuccessful in the training set. A soft margin is provided in order to address this issue. It is frequently applied to behaviour-based classification techniques and classification problems. Usually, it offers an original answer. This method, which is based on decision-making trees, has gained popularity recently. In order to arrive at a final prediction, the training demonstrates the repetitive addition of new trees that predict residuals or mistakes of earlier trees that are compared with earlier trees. Transactions are categorized as either legal or fraudulent by this system [7]. The SMOTE approach was used to accomplish oversampling, which is then maintained in the XGBoost classifier. The aforementioned algorithms were created to accurately forecast fraudulent transactions.

IV. PROPOSED SYSTEM

We evaluate the XGBoost algorithm against the system's hybridize CNN-SVM paradigm. The suggested hybrid approach combines the salient features from the two classification algorithms. In the proposed hybrid method, CNN serves as an automatic feature extractor and SVM as a two-dimensional predictor and suggested model's approach is trained and tested on the handwritten digit's dataset from MNIST. XGBoost enhances the outcomes by using a more comprehensive approach to control over fitting. In order to select the boost round for each run, cross-validation features are included, and it can manage missing data. To get better results, XGBoost has to have a few parameters adjusted. To obtain the findings in this experiment, the XGBoost approach's parameters are adjusted as follows.

A. Hybrid CNN-SVM:

An SVM (Support Vector Machine) classifier was used to replace the CNN (Convolutional Neural Network) model's final output layer in order to create the CNN-SVM. The first classifier is from beginning to end developed layer that is fully linked with Soft-max layer. The final completely linked and Soft-max layers are deleted to create an SVM on top of the first classifier. Here the CNN is a convolutional neural network that contains various layers that has a deep learning (supervised) architecture, primarily made up of four

layers. The components required for handling one-dimensional data., there are layers of convolution, layers of pool dropout layers, and activation functions This approach utilizes a combination of CNN and SVM to form a new Hybrid model where the features are combined to get the better result [9].

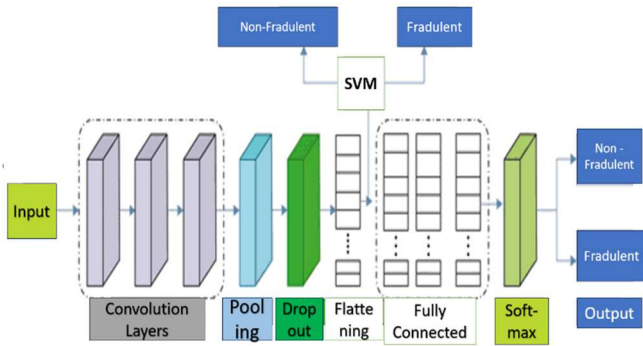


Fig. 1. Architecture model of hybrid

a) CNN-SVM

In fig:1 The hybrid architecture of CNN-SVM was developed by replacing the final layer of the CNN framework with an SVM classifier. SVM was applied after networks with a soft-max function that were fully connected. A binary SVM classifier is built on over of the softmax layer, which is eliminated until the final layer. The convolutional layer differentiates the data based on the characteristics of transaction information, and the pooling layer shrinks the feature map sizes [11]. We used the dropout strategy, which, in order to keep the system from excessive fitting, eliminates 50 percent of the layers within the period of training following three consecutive convolutional layers. It lowers the computational cost with the aid of the max pool layer. The outcome is then given a fattening layer, which reduces the three-dimensional vector to one dimension. Next, a target variable was prognosticated using three small layers with SoftMax and Relerning functions for activation. When it comes to binary classification problems, accuracy and F1 score are the most crucial analyses. When the target class is in balance and not out of balance, accuracy can be useful.

b) Algorithm Hybrid Model

Step1: Import the information that was provided in.
 Step2: preparing the data in advance and eliminating fraudulent transactions.
 Step3: Utilizing the chosen CNN feature.
 Step4: SVM classifiers are used.
 Step5: Model of prediction.

B. XGBoost

XGBoost is a decentralized gradient optimization toolkit designed for machine learning model training that prioritizes performance and flexibility. It is an ensemble learning method that combines the predictions of multiple inadequate models to achieve a more robust result. One of the main features of XGBoost is its ability to effectively handle

missing values from real-life without needing extensive preparation. Additionally, XGBoost has integrated multitasking capability, allowing quick model training on large datasets. The XGBoost predictor can be optimized with Hyperopt to create a model that has ideal characteristics and strong fraud detection performance. We deploy the XGBoost model on the online transaction fraud detection system to provide good detection performance. benefits in terms of application and operating time, as well as the capacity to precisely forecast the likelihood of fraud in network transaction behaviours.

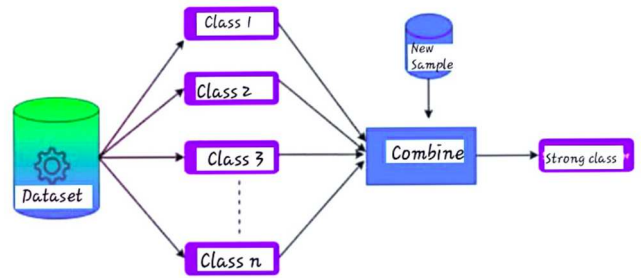


Fig. 2. Architecture of XGBoost

a) XGBoost-Based Fraud Detection Algorithm

XGBoost steps include pre-processing the data, fine-tuning the parameters, and building the model.

- 1) **Preparing the data:** The basis of additional analysis is provided by activity as well as identification information collections. For clear features, utilize indicator encoding as a method to transform discrete data into [13-15] numerical values that range from 0 to N-1. This process allows us to represent categorical variables in a format suitable for various analytical techniques, ensuring that each category is distinctly represented by a unique numeric value.
- 2) **Parameter tuning:** Hyperopt is used to fine-tune the model's parameters following data pre-processing. Hyperopt can implement XGBoost classifier tuning parameters rapidly and effectively.
- 3) Building the final model requires create the classification system XGBoost using the best settings then retrain it on all available data.

b) Algorithm

Step1: Design a deliberation characteristic.
 Step2: Create an area for infrastructure.
 Step3: Training model.

C. Model performance evaluation techniques

Reliability, relevance, efficiency, AUC, and F1-score are used for measuring model performance. Efficiency is a measure used to assess how often a machine learning model

accurately predicts a result. To calculate accuracy, divide the number of correct predictions by the total number of predictions.

$Accuracy = (TP + TN) / (TP + TN + FP + FN)$ (1)
This methodology allows for four different outcomes when evaluating the efficacy for detecting frauds and creating model with more prediction.

- i. True positives (TP): The forecasts of the model turned out accurate. In that scenario, a consumer not dishonest, and model makes the correct prediction.
- ii. True negatives (TN): Here, the customer lacks non-fraudulent data, but the framework fails to forecast it; in other words, When the model indicates that nothing has happened, it actually means that something has occurred.
- iii. False positives (FP): If the system determines this anything happened, the model predicts fraud even if the consumer has no non-fraudulent evidence. Those are classified as kind I mistakes.
- iv. False negatives (FN): Whenever the simulation is unable to provide any guarantees because the event has already taken place, for as when a consumer is aken advantage of fraudulent which the system failed to predict. That represents a deadly error, also known as a kind II mistake.

Reliability is a measure that quantifies the percentage of accurately predicted beneficial outcomes out of all possible beneficial outcomes. It is calculated by dividing the number of correct positive predictions (true positives) by the total number of positive instances predicted by the model. (Including true and false positives).

$$Precision = TP / (TP + FP) \quad (2)$$

Relevance is a measure that measures how frequently a machine learning model correctly finds instances that are favourable (which is correct benefits) out of every one of the real positive data points in a set of data. To compute recall, divide the number of true positives by the number of positive cases. This category includes both genuine positives (instances that were correctly identified) and erroneous negatives (such as those that were ignored).

$$Relevance = \frac{TP}{f_1} (TP + TN) \quad (3)$$

The F1-scoring function combines recall and precision. The F1- score is the weighted average of precision and recall. Therefore, both false positives and false negatives are considered while generating this rating. Equation 1-4 frequently outperforms precision, particularly if the classes are unevenly distributed.

$$F1 - score = 2TP / 2TP + FP + FN \quad (4)$$

The Receiver Operating Characteristic (ROC) curve, or AUC-ROC, is a probability curve that compares the true positive rate (TPR) and the false positive rate (FPR) at various thresholds to differentiate between "transaction data" and "fraudulent data. AUC metrics can discriminate between fraudulent and non-fraudulent classes.

V. RESULT ANALYSIS

TABLE I. SHOWS EVALUATION OF PERFORMANCE D = 0.01, GAMMA = 0.09.

Proposed methods	Reliability	Relevance	Efficiency	F1-score	AUC
CNN-SVM (HYBRID)	91.08	90.50	90.34	90.41	91.05
XGBoost	89.74	88.60	76.90	82.34	89.49

TABLE II. EVOLUTION PERFORMANCE D = 100; GAMMA = 0.09.

Proposed methods	Accuracy	Precision	Recall	F1-score	AUC
Rbf	91.85	90.51	90.3	90.4	91.0
Linear	90.61	94.40	87.29	90.90	90.73
Sigmoid	78.15	62.81	79.00	70.00	73.80
Polynomial	73.55	67.43	76.31	73.48	71.60

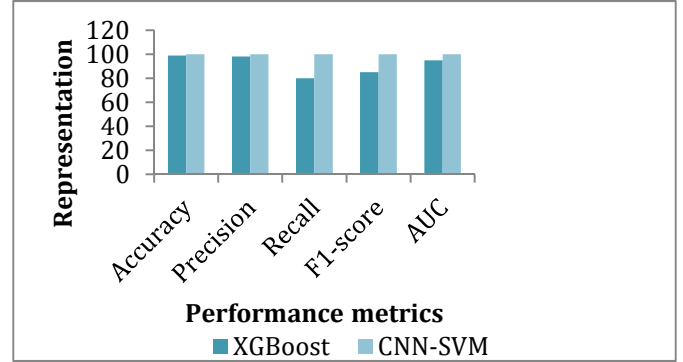


Fig. 3. Performance metrics are assessed by comparing a hybrid model that combines CNN and SVM with the XGBoost algorithm.

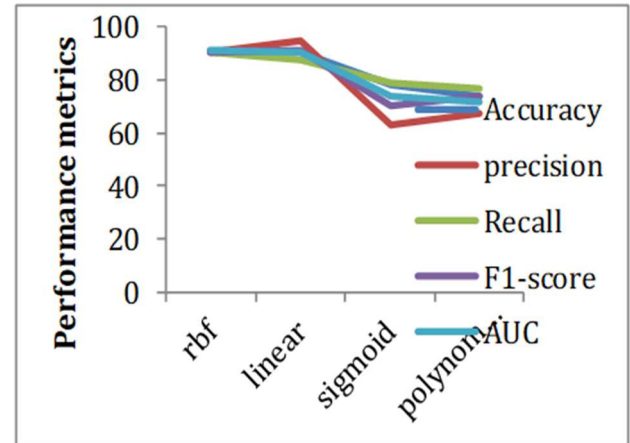


Fig. 4. Comparing hybrid CNN-SVM with various performance metrics.

In the first experiment, a convolution neural network extracted attributes, which were then classified using a network that was completely interconnected with Soft-max. The second test employed a support vector machine (SVM) classifier. The SVM was configured with a regularization parameter C set to 0.01, a gamma value of 0.09, and utilized

a radial basis function (RBF) kernel applied to the flattened layer. This setup achieved a baseline efficiency of 91.08%.

Hybridized CNN-SVM performance was compared with the XGBoost algorithm in fig:4. As a result, we may conclude that our model predicts fraud detection fairly well. According to the experimental results, our suggested CNN-SVM algorithms outperform XGBoost in terms of fraud detection reliability, relevance, F1-score, and area under the curve (AUC) of 90.50%, 90.34%, 90.41%, and 91.05%, respectively in fig:3. As a result, precision, recall, F1-score, and AUC are useful indicators for determining how effectively the fraud detection model is doing.

VI. CONCLUSION

Now a day's atmosphere is rife with fraud because most of us regularly use debit cards. Crime has increased as an outcome of the rise in transactions over the Internet and considerable financial losses. This is due to technological developments also online mode transactions, which have resulted in fraud and large losses. Methods that are effective as well as efficient are needed to detect fraud in debit card transactions. Processes that are both efficient and effective are needed to identify fraudulent in digital transactions. In this research, we evaluated the effectiveness of XGBoost and the Hybrid CNN-SVM model for identifying fraudulent credit card transactions. We evaluated the effectiveness of our suggested hybrid CNN-SVM model for fraud detection using actual, accessible-to-everyone transactions on the internet data.

The dataset available at <https://www.kaggle.com> comprises an overall of 24,000 transactions. Thirty features total in the dataset: amount, time, H1, ..., H28. The objective variable has a significant degree of skewness and includes both fraudulent and non-fraudulent transactions. Therefore, algorithms can forecast subsequent transactions that are legitimate with accuracy; they are unable to accurately foresee future operations that are false. The experimental results reveal that our suggested strategy beats the XGBoost method in terms of reliability, relevance, F1-score, and AUC when predicting whether a card transaction is fraudulent or not. Finally, our proposed hybridization strategy outperforms existing methods in terms of fraud prediction. The hybridization

strategy outperforms other suitable models in terms of fraud detection, which is one of our method's primary advantages.

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